
Understanding where physical theories fail without knowing what replaces them

Chenxi He*

Cavendish Laboratory
University of Cambridge
Cambridge CB3 0HE, United Kingdom

Jingqiu Chen

College of Physics and Optoelectronic Engineering
Shenzhen University
Shenzhen 518060, China

Abstract

Physical predictions can fail before a theory explains the failure. We ask how to test such a failure while restricting the knowledge behind the prediction. We make the historically admissible model class an explicit input to a calibrated test: a calendar date, the knowledge horizon, fixes which physical laws may construct the prediction, and later theories enter only for retrospective comparison. Measurements are compared with a named prediction under stated measurement errors, and generic screens, needing no named prediction, rank relations by how unlike the admissible laws they are. At an 1899 horizon, all five pre-1900 predictions are rejected on five selected historical datasets. On the same grids and error models, four of the five historical successors are not rejected, whereas retrospectively chosen alternatives with the wrong shape or mechanism remain rejected. A test rejects the classical prediction for Millikan’s nearly perfectly linear photoelectric data: curve familiarity is not physical compatibility. A neural encoder trained only on pre-1900 relations ranks post-1900 laws above pre-1900 controls, with performance depending on how the reference formulas were collected and generated. The historical constraint allows the predictive domain of specified physical models to be tested without using a successor theory to construct the test.

1 Introduction

The failure of a physical prediction can be evident before its explanation is known. Classical predictions for radiation, heat capacity and electrical resistance provide familiar examples: measurements outside a successful regime eventually required quantum descriptions. A test of predictive failure, however, requires a reference prediction and an account of the measurement errors. The question is how to construct that reference while excluding the knowledge that later explained the departure.

Existing methods quantify discrepancy, detect change or search for replacement laws. Bayesian model discrepancy compares observations with a simulator [1], conformal prediction calibrates residuals [2], change-point methods locate distributional shifts [3] and out-of-distribution detectors score departures from a reference distribution [4]. Machine-

*Correspondence: ch2067@cam.ac.uk

learning methods interpolate, forecast or extrapolate scientific data [5, 6], and symbolic regression searches for governing expressions [7, 8]. Each takes the reference model or the space of candidate laws as given. None specifies which physical models were admissible before the later explanation existed. For machine learning the gap is a knowledge cutoff that is never stated: a model trained on all recorded formulas cannot show what was knowable before a discovery.

We make that specification an explicit input. A calendar date, the knowledge horizon, fixes the knowledge a test may use: a relation is admissible if its physical content follows from laws and principles documented before the horizon. The pre-horizon prediction for a measured quantity is built from admissible relations only. The historical successor that later explained a departure enters afterward, for comparison on the same measurements, so it never defines the discrepancy being tested.

EPOCH (Epistemic Probe of Classical Horizons) separates generic screens from tests of named predictions. A representation screen measures how far a curve lies from pre-horizon relations in the embedding (the vector representation a neural encoder assigns to each curve) of an encoder trained only on those relations. A simple-form extrapolation screen, which fits power, exponential, linear or constant forms, fits the best form over the stretch of data it describes, the fitted regime, and scores its extrapolation over the remainder, the extrapolation region. Both screens identify unfamiliar curves; neither refers to a physical theory. A named-prediction test compares the residuals of the measurements from a specified pre-horizon prediction with a null distribution simulated under the stated measurement errors. It returns one of four verdicts: compatibility over the measured range; localized incompatibility, when the prediction fits a regime and fails beyond an estimated boundary; whole-range incompatibility, when the prediction is rejected over the sampled range and no compatible regime is identified; or abstention, when the inputs do not support a test.

We use an 1899 horizon, for which classical relations can be reconstructed from dated sources. We first show how the horizon defines the admissible prediction for a dataset. We then test five historical predictions on measurements and replace each by its successor on the same grid and error model. Next we ask how far the generic screens rank later laws above pre-1900 laws and where they stop being informative. Finally we show that the observation model determines both rejection and the localization of a boundary.

2 Results

2.1 A historical cutoff defines the admissible prediction

Figure 1a places dated law families on either side of the 1899 horizon. Relations sharing a law form a law family, and training and evaluation are divided by family. Pre-1900 families supply both the relations on which the representation screen is trained and the predictions a test may use; later families enter only for evaluation. A test (Fig. 1b) compares measurements with the admissible pre-1900 prediction under an observation model, the stated description of the errors, censoring and grid, and returns a verdict. The generic screens form a side branch that flags relations for testing (Fig. 1b,c). They receive only a point cloud, the normalized (x, y) pairs of a relation without formula, units or labels. Where a fitted regime is sampled, the extrapolation screen can also localize a departure, which gives a descriptive classification; rejection of a named physical prediction comes from the named-prediction test. A fixed rule assigns each dataset to the tests its inputs support (Methods; Fig. S1 of the Supplemental Material [9]). Figure 1d shows the verdicts on real data, and Table 1 summarizes the evidence.

2.2 Historical predictions fail where their successors fit

All five pre-1900 predictions are rejected, and on the same grids and error models four of the five successor predictions are not rejected (Fig. 2). We selected these measurements as known classical failures. The cosmic microwave background spectrum from the Cosmic Background Explorer Far Infrared Absolute Spectrophotometer (COBE-FIRAS) and Bertozzi's electron data lack a measured classical regime, whereas five heat-capacity curves and mercury

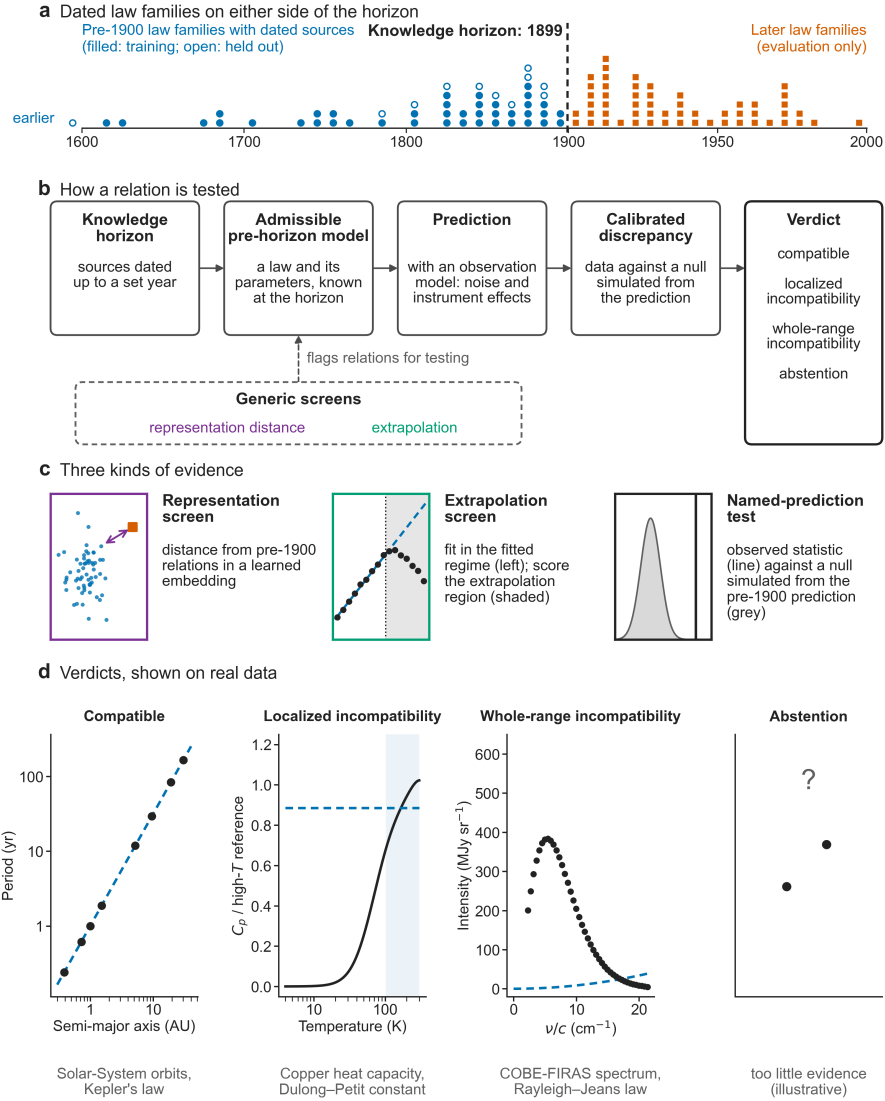


Figure 1. Testing a prediction under a knowledge horizon. **a**, Pre-1900 law families with dated sources (blue circles; filled, used to train the compact encoder; open, held out) and the later law families of the benchmark (vermillion squares), one point per family at its first dated source. The time axis changes scale at 1900. The open circle marked ‘earlier’ is a held-out family dated about 250 BC. **b**, How a relation is tested. The knowledge horizon (sources dated up to a set year) fixes the admissible pre-horizon model, a law and its parameters known at the horizon; its prediction, with an observation model for noise and instrument effects, gives a discrepancy calibrated against a null simulated from the prediction, and the test returns one of four verdicts. The generic screens (dashed box; representation distance and extrapolation) form a side branch that flags relations for testing; where a fitted regime is sampled, extrapolation can also localize a departure. **c**, Schematic of the three kinds of evidence: the distance of a new point cloud (normalized (x, y) pairs without formula or units; vermillion square) from pre-1900 relations in the encoder’s embedding (representation screen); a form fitted in the fitted regime (solid blue) and extrapolated to the extrapolation region (dashed, gray shading), with measurements in black (extrapolation screen); and an observed statistic (black line) against a null distribution simulated from the pre-1900 prediction (gray) (named-prediction test). **d**, Verdicts on real data, with pre-1900 predictions dashed: Solar-System orbits under Kepler’s law, a classical control (compatible); the National Institute of Standards and Technology (NIST) copper heat-capacity curve under the Dulong–Petit constant fitted in the shaded fitted regime (candidate localized incompatibility; normalized as in Fig. 2c); the Cosmic Background Explorer Far Infrared Absolute Spectrophotometer (COBE-FIRAS) spectrum under the fitted Rayleigh–Jeans law (whole-range incompatibility). The abstention panel is illustrative.

Table 1. Summary of the evidence. Each row gives one headline result. AUROC, the area under the receiver operating characteristic curve (0.5 is chance), and the same-generator set (a separate set of later formulas produced by the training corpus’s generator) are defined in the text. Intervals are 95% bootstrap intervals over families or mechanism groups. Every quantitative comparison and its evaluation population is listed in Note S19.

Evidence	Headline result	Interpretation
Historical prediction tests, five selected datasets	All five pre-1900 predictions rejected; four of five successors not rejected on the same grid and error model	A stated prediction can be tested without using its successor to construct the test
Generic-screen ranking, generated formulas	Large encoder ranks 55 later families above nine pre-1900 control groups with AUROC 0.925 [0.851, 0.982], and the same-generator set at 0.536 [0.424, 0.643]	Ranking of later relations depends on the formula corpus and its generator
Measured transfer, sixteen series in 5 against 9 groups	Compact encoder 0.62 [0.29, 0.89]; extrapolation screen 0.78 [0.38, 1.00]; both intervals span chance	Transfer to measurements not demonstrated
Detection and observation effects, synthetic	True-positive rate 0.271 [0.073, 0.500] at the 5%-targeted threshold (0/96 false positives); false-positive rates 0.40 under saturation and 0.63 under censoring at the highest simulated severity	Calibrated detection is weaker than ranking; departures need an observation model

83 resistance contain one, so the cases distinguish failure across the sampled range from failure
84 beyond a fitted regime (Fig. 2a–d). Millikan’s photoelectric line tests a different distinction:
85 a familiar mathematical shape can violate a named physical prediction.

86 The COBE-FIRAS monopole spectrum of the cosmic microwave background [10] has 43
87 points and, at $T = 2.725$ K, samples $x = h\nu/kT$, with h and k the Planck and Boltzmann
88 constants, from 1.2 to 11.3. The complete range lies outside $x \ll 1$, where the Rayleigh–Jeans
89 law [11, 12] holds. Its prediction, intensity proportional to ν^2 , over-predicts the intensity
90 above the peak by orders of magnitude, by an amount that depends on the fitting window
91 (Note S5). The verdict is whole-range incompatibility; the data do not locate a transition
92 from a classical regime.

93 Bertozzi’s electron measurements [13] likewise lie outside the non-relativistic regime. For
94 kinetic energy against $\beta^2 = v^2/c^2$, the classical law $\frac{1}{2}mv^2$ crosses $\beta^2 = 1$ at 0.26 MeV and
95 reaches $\beta^2 = 58.7$ at 15 MeV, equivalent to $7.7c$. The lowest measured point is already $0.87c$,
96 and the measured β^2 values approach 1. The verdict is whole-range incompatibility with the
97 Newtonian energy–velocity prediction.

98 The Dulong–Petit law [14] predicts temperature-independent molar heat capacity. Five of
99 the six National Institute of Standards and Technology (NIST) cryogenic reference curves
100 contain a high-temperature plateau, the fitted regime, and depart from that constant at low
101 temperature, giving a candidate localized incompatibility; beryllium lacks a plateau and gives
102 whole-range incompatibility. For oxygen-free high-conductivity copper, the extrapolated
103 plateau over-predicts the 4 K reference value by 3.7×10^3 . The six reference curves share a
104 low-temperature mechanism and are grouped together.

105 Kamerlingh Onnes’s mercury resistance values [15] contain a smooth normal-state trend
106 followed by a discontinuous drop. The transition is placed at 4.20 K, a temperature fixed
107 from the historical record in the test that pairs the pre-1900 prediction with its successor.
108 The smooth normal-state and residual-resistance models [16] are tested with figure-reading
109 uncertainty and the reported resistance bound below $10^{-5} \Omega$ in the observation model.

110 Millikan’s 1916 measurement [17] demonstrates that curve familiarity is not physical compat-
111 ibility. Stopping potential is linear in frequency, $V_{\text{stop}} = (h/e)(\nu - \nu_0)$, with $R^2 = 0.99995$
112 and a slope recovering h/e to 0.3% (Fig. 2e). The extrapolation screen accepts the shape, yet
113 the named-prediction test on the same four measurements rejects the classical prediction of

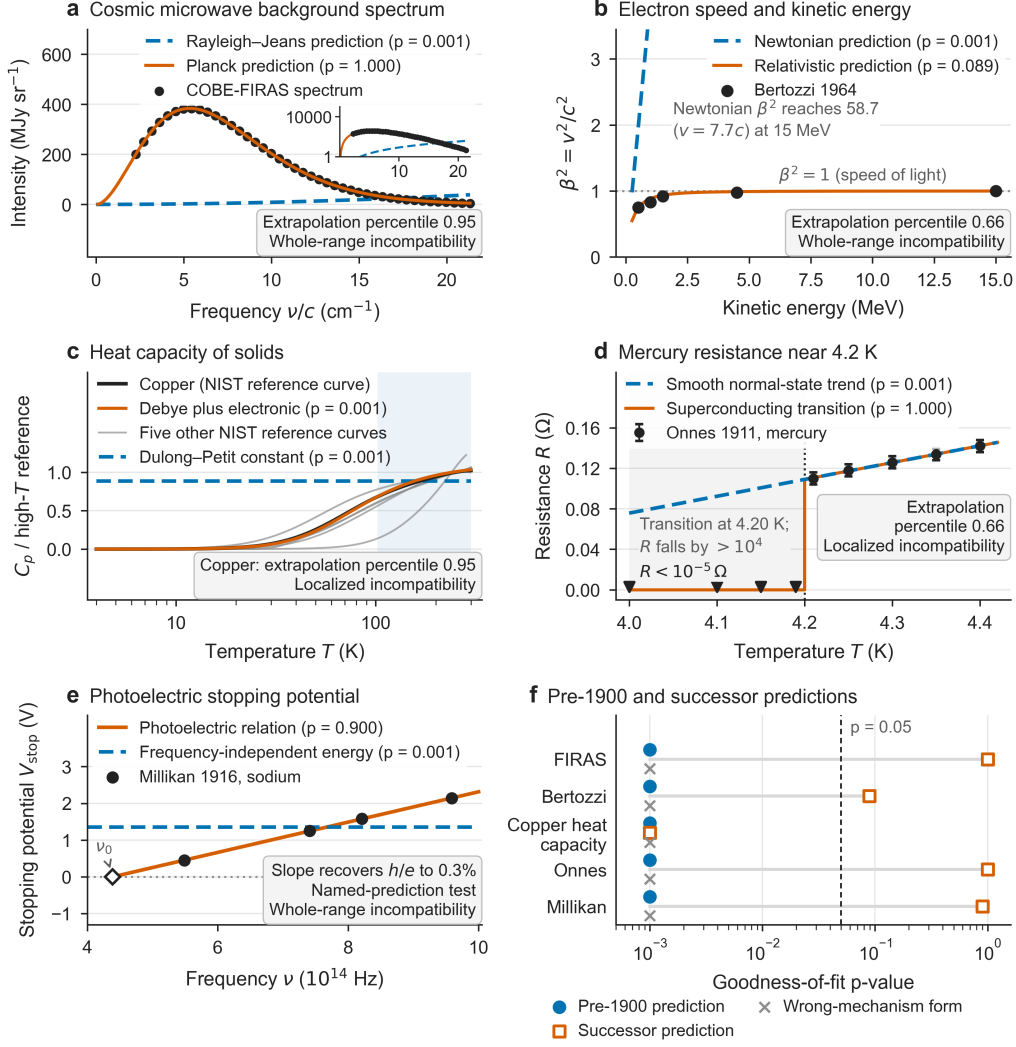


Figure 2. Historical prediction tests under the 1899 horizon. **a–d**, Black points are published processed measurements or digitized historical values; in **c** the black curve is the NIST reference curve for copper and the gray curves are the five other solids. Blue dashed curves are the pre-1900 predictions as fitted in the goodness-of-fit test (for mercury, the smooth normal-state trend; for copper, the constant fitted over the shaded fitted regime) and vermilion curves the successor predictions fitted on the same grid (Planck, relativistic kinetic energy, Debye plus electronic term and a superconducting transition); legends give the goodness-of-fit p -value of each named prediction (also in **f**), not a test of boundary localization. Gray boxes give the extrapolation-screen percentile, the empirical percentile of the extrapolation score among the size-matched classical calibration scores (upper-tail p -values in Note S4), and the verdict; the localized verdicts in **c** and **d** are descriptive candidate classifications, and in **e** the box gives the slope check and the test giving the verdict. The inset in **a** repeats the spectrum on a logarithmic intensity scale. FIRAS and Bertozzi yield whole-range incompatibility. Specific heat and Kamerlingh Onnes contain a fitted regime and a candidate boundary. In **c** heat capacity is normalized by its median above 70% of the maximum temperature. Shading marks the fitted regime in **c** and the region beyond the transition in **d**; in **d** the dotted line (labeled 4.20 K) marks the historically recorded transition temperature, error bars give the figure-reading uncertainty and downward triangles mark the four values below the reported limit of $10^{-5} \Omega$. **e**, In Millikan’s 1916 digitized values, stopping potential rises linearly with frequency as $V_{\text{stop}} = (h/e)(\nu - \nu_0)$. The photoelectric relation (vermilion) recovers h/e to 0.3%, and the open diamond marks the extrapolated zero crossing ν_0 . The extrapolation score falls below the threshold, whereas the named-prediction test rejects the classical prediction of frequency-independent energy (blue dashed line). **f**, Goodness-of-fit p -values on the same measurement grid under the pre-1900 prediction (filled blue circles), its successor (open vermilion squares) and a wrong-mechanism form (gray crosses), joined by a gray line for each dataset; the vertical dashed line marks $p = 0.05$.

frequency-independent photoelectron energy. The fitted intercept gives $\nu_0 = 4.39 \times 10^{14}$ Hz; all four frequencies exceed ν_0 , so the verdict is whole-range incompatibility.

Each pre-1900 prediction was then replaced by its historical successor and scored on identical measurements and grids under the same error model (Fig. 2f); successor parameters not fixed by theory (nuisance parameters, such as amplitude) are fitted to the same data where specified, and the Kamerlingh Onnes successor is a phenomenological transition form. All pre-1900 predictions are rejected (bootstrap $p = 0.001$). The successors are not rejected for FIRAS under Planck’s law [18], Bertozzi under relativistic kinetic energy [19], Kamerlingh Onnes under a superconducting transition model and Millikan under the photoelectric relation [20]; individual p -values are given in Fig. 2 and Note S11. The copper successor, a Debye-plus-electronic form [21], is rejected at the primary 5% error floor (a minimum relative error assigned to the reference curve) ($p = 0.001$) and passes when the floor is raised to 7.5%; the floor stands in for a covariance model of the reference curve, which is unavailable.

Wrong-shape and wrong-mechanism alternatives, one of each per case and retrospectively chosen after the datasets were known, remain rejected: all five wrong-shape alternatives after Benjamini–Hochberg false-discovery-rate correction [22] (adjusted $q \leq 0.0015$) and all five dimensionally coherent wrong-mechanism alternatives (1,000 bootstrap replicates; $p = q = 1/1001$). The tested alternative forms therefore do not reproduce the successors’ agreement (Note S11).

Measured classical controls retain their expected forms: Solar-System orbits [23] and the four Galilean moons recover Kepler’s exponent of 3/2 to four figures (1.500 and 1.500), and Boyle’s 1662 pressure–volume table [24] gives -0.998 ; all three fall below the extrapolation screen’s 0.5 screening threshold (percentiles 0.02, 0.00 and 0.38; Fig. S2a–c). The extrapolation screen does not reject FIRAS, copper heat capacity, Bertozzi or Kamerlingh Onnes after Benjamini–Hochberg correction (Note S4), so the rejections come from the named-prediction tests, and the localized verdicts for heat capacity and resistance are descriptive classifications.

2.3 Generic screens rank later laws but do not replace a test of a physical prediction

A pre-1900 representation ranks later laws, with performance that depends on how the reference formulas were collected and generated. The large encoder, trained only on point clouds from dimensionless reductions of pre-1900 laws, ranks 55 later law families above nine held-out pre-1900 control groups with AUROC 0.925 [0.851, 0.982] (Fig. 3a). AUROC, the area under the receiver operating characteristic curve, is the probability that a randomly chosen later family outranks a pre-1900 group, counting ties as one half; 0.5 is chance. Brackets give 95% bootstrap intervals over families or groups. The untrained network, the same network with its initial random weights, gives 0.649 on the same comparison, and each of the encoder’s four distance components improves on it (Fig. S4d).

The ranking depends on how the reference formulas were collected and generated. Regenerating the nine control groups with the later families’ sampling and noise still gives AUROC 0.857 [0.741, 0.945], against 0.545 for the untrained network (Fig. 3a); these regenerated controls were added after the primary results were known. The dependence appears on the same-generator set, a separate collection of later formulas dated 1901–1950 and produced by the generator of the training corpus, whose family composition also differs from the 55-family benchmark: there the trained encoder gives 0.536 [0.424, 0.643], near chance. The learned ranking therefore reflects which formulas were collected and how they were generated, as well as when the laws were found. An exploratory extension to a 1950 horizon, whose added formulas lack complete source-by-source dates, gives 0.453 [0.228, 0.678] on post-1950 families (Note S16).

Simple-form extrapolation also ranks the later families above pre-1900 controls, without a trained network; on synthetic benchmarks it reaches AUROC 0.80 [0.64, 0.94], and its extrapolation-residual ratio (0.97–1.00) exceeds standard misspecification and change-point tests applied to the same classical fit (0.73 and 0.67; Fig. S2d–f). Against eleven cited pre-1900 families excluded from encoder training, the extrapolation screen gives AUROC 0.888 [0.800, 0.959], compared with 0.669 [0.534, 0.798] for a compact encoder trained on 205 point clouds from 40 source-verified pre-1900 families (Fig. 3b). The compact encoder

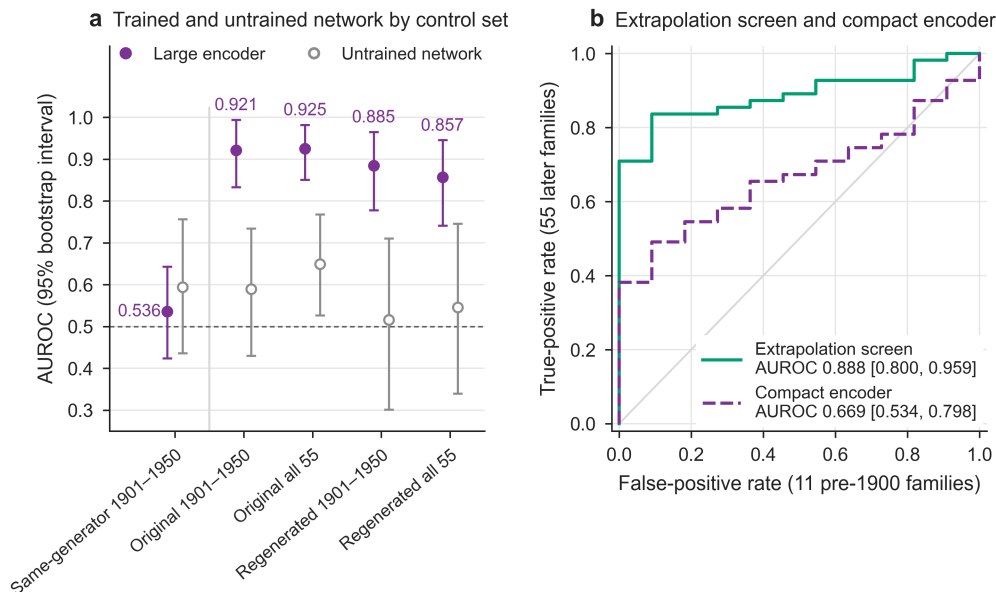


Figure 3. Generic screening at the 1899 horizon. **a**, AUROCs of the large encoder (filled purple, values labeled) and the untrained network (open gray) for the same-generator set and for each combination of later families (1901–1950 or all 55) and pre-1900 controls (original or regenerated), with 95% bootstrap intervals over families or groups; the same-generator comparison is shown left of the vertical line and the dashed line marks chance. **b**, Receiver operating characteristic (ROC) curves of the extrapolation screen (green) and the compact encoder (purple, dashed) for the 55 later families against 11 cited pre-1900 families excluded from encoder training; the diagonal marks chance. Learned score distributions, the compact encoder’s controls and the large encoder’s ROC curves and distance components are shown in Fig. S4.

ranks 12 synthetic departure families, curves built to leave a classical form, above 12 classical families with family AUROC 0.903, against 0.375 for an encoder matched in architecture, training and data size but trained on generic smooth expressions (Fig. S4c; Note S8). Ranking is stronger than calibrated detection: at a threshold targeted to 5% false positives, the extrapolation screen detects synthetic departures with true-positive rate 0.271 [0.073, 0.500] (Note S4), and at their calibrated thresholds each screen flags 7 of the 55 later families and none of the 11 controls (Note S15).

The families the screens miss show why a ranking does not test a physical prediction. Threshold and piecewise relations separate more strongly by extrapolation, whereas affine, power and exponential relations can remain close to the admitted forms. Later laws with familiar shapes, such as special relativity, the photoelectric threshold and radioactive decay, are flagged by neither screen and require a named-prediction test (Note S15). Eleven further generated cases, assembled after the primary scores were known, illustrate the same distinction (Fig. S5; Note S17). On sixteen measured NIST series, grouped into five post-1900 and nine classical or instrument-calibration (metrological) mechanism groups and collected during method development, the compact encoder gives AUROC 0.62 [0.29, 0.89] and the extrapolation screen 0.78 [0.38, 1.00]; both intervals span chance (Fig. S6; Note S18).

2.4 Observation models determine rejection and localization

The observation process can produce departures even when the underlying classical relation is unchanged. At the highest simulated severity, false-positive rates of the extrapolation screen are 40% for saturation and 63% for censoring under calibration targeted to 5%; the other tested effects remain at or below 11% (Fig. 4a). A plateau or threshold therefore requires an observation model that accounts for these effects, as the dependence of the copper comparison on its error floor also shows.

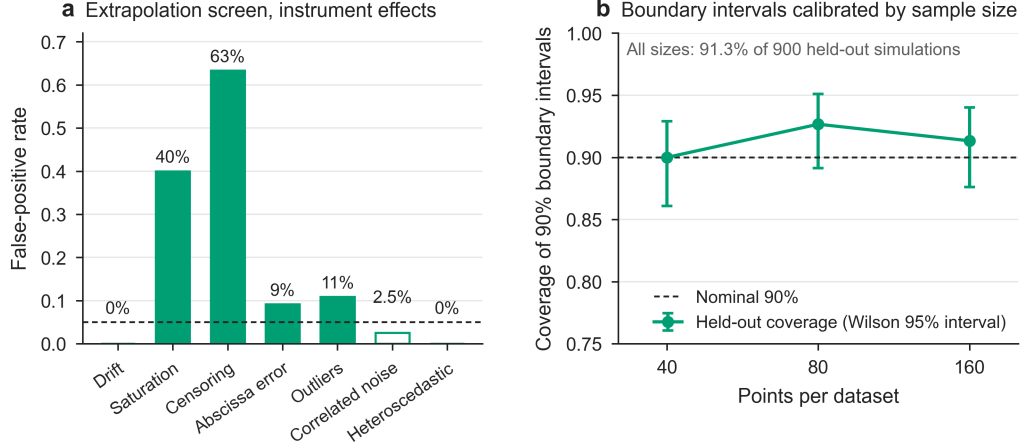


Figure 4. Observation effects and boundary calibration. **a**, False-positive rates of the extrapolation screen under simulated instrument effects (abscissa error: noise in the independent variable) at the highest severity tested, with calibration targeted to 5%; the dashed line marks the nominal rate, open bars are at or below it and rates of 0% have no visible bar. **b**, Held-out coverage of 90% boundary intervals calibrated separately at each sample size (40, 80 and 160 points); points give coverage, bars give Wilson binomial 95% intervals and the dashed line marks nominal coverage.

193 Localizing a boundary requires intervals calibrated beyond ordinary measurement resampling.
 194 When a compatible regime and its departure are both sampled, a nominal 90% measurement-
 195 bootstrap interval covers 53.9% of simulated boundaries (358 simulations of steps, saturation,
 196 kinks, smooth turnovers and crossovers) and 8.5% of steps. We calibrated the interval radii
 197 on 899 independent simulations, separately by sample size, raising held-out coverage to 91.3%
 198 [89.3%, 93.0%] across 900 simulations (Wilson binomial interval; Fig. 4b). This coverage is
 199 averaged over the simulated boundaries (marginal), not guaranteed for each boundary type
 200 (Note S13).

201 3 Discussion

202 Detecting an unusual curve is not the same as testing a physical prediction. A screen ranks a
 203 curve against a reference corpus; only a named prediction, stated with its observation model,
 204 can be rejected. A historical knowledge constraint makes that test possible for predictions
 205 whose failure was later explained, without letting the successor define the discrepancy: the
 206 reference is built from pre-horizon physics, and the successor enters afterward on the same
 207 grid and error model. In four of the five historical comparisons this replacement turns
 208 rejection into non-rejection, whereas the tested alternative forms keep their discrepancies.

209 Millikan’s line and a classical interference resonance show why the distinction matters. Mil-
 210 likan’s line has an ordinary shape, yet its frequency dependence rejects the classical prediction
 211 (Fig. 2e). Conversely, a sharp interference resonance in a measured transmittance series
 212 follows from classical wave optics; its learned score is low (0.063) although its extrapolation
 213 score (1.95) exceeds the classical median (0.57) (Fig. S7). Extrapolation of a fitted generic
 214 form tests a shape, and the representation screen measures familiarity; a claim about a
 215 theory needs that theory’s prediction.

216 EPOCH combines neural representation methods [5, 6], calibrated residuals and model-
 217 discrepancy testing [2, 3, 1]. It is related to model-independent searches in particle physics [25]
 218 and to machine rediscovery of laws [7, 8, 26]. Symbolic regression can seek candidate
 219 replacements under a parsimony prior [27]; testing a restricted prediction and searching
 220 for its replacement address different questions. The historical constraint specifies which
 221 knowledge can enter the reference prediction.

The evidence has limits. Because the measurements were selected as known failures, the historical comparisons demonstrate the construction but do not give a prospective detection rate. Rejection and localization depend on the observation model, as copper’s error-floor sensitivity and the saturation and censoring simulations show. The learned ranking depends on the formula corpus and its generator, and transfer of the screens to measured series and to a later horizon has not been demonstrated. On measured data the largest learned scores belong to a standard-resistor record and a polymer-impedance curve, so a high learned score marks an unusual relation, not a physical mechanism.

A prospective study would pair independently selected measurements with a dated corpus, specified pre-horizon predictions and externally supported error models. With a simulator-based reference, the same specification extends to physical descriptions that supply predictions without a closed-form law, and screening can direct attention to relations that a named-prediction test then examines. Making the admissible knowledge explicit allows the predictive domain of a historical model class to be tested without the theory that replaced it.

4 Methods

4.1 Historical admissibility

A relation is admissible at the knowledge horizon, here 1899, if its physical content follows from laws and principles documented on or before that date. The date is set by the source law, not by when a particular formula text was printed. Mathematical rewriting that adds no physical content, such as nondimensionalization, reduction to one variable or algebraic rearrangement, is allowed. The Rayleigh–Jeans form follows from the equipartition theorem applied to the electromagnetic modes of a cavity, and the normal-state resistance model from Matthiessen’s 1864 decomposition of resistance into a temperature-dependent part and a constant residual part [28]; both are therefore admissible pre-1900 predictions. Where the standard statement of a prediction was published later, as for the Rayleigh–Jeans form in 1900 and 1905 and the Drude model in 1900, we cite that statement as the reference and do not use it as a training source. Only admissible relations enter the training corpora and the predictions a test may fit.

Relations derived from one physical law traced to one dated source form a law family, and measured series produced by one physical mechanism form a mechanism group. Both are the units over which AUROCs are computed and bootstrap intervals resampled; the nine pre-1900 control groups are sets of law families that share one mechanism.

4.2 Specifying a test

An EPOCH test specifies $\mathcal{T} = (\mathcal{M}, \Theta, P_\varepsilon, I, A, S, \alpha)$. Here $\mathcal{M}$ lists the admissible models, Θ the fitted parameters, P_ε the observation model, I the fitted regime, A every adaptive choice, S the discrepancy statistic and α the error rate. The observation model comprises the error distribution, censoring or saturation, and the grid; where no published covariance exists, an error floor assigns a minimum relative error. Direction choice, window search, model selection and parameter fitting all belong to A and are rerun inside every calibration and null replicate.

The named-prediction test computes the residuals of the measurements from the pre-horizon prediction, scaled by the stated errors, and compares their size with a null distribution: the distribution of the statistic over datasets simulated from that prediction on the same x -grid with the same error model. Historical named-prediction p -values use parametric bootstrap simulations under the specified observation model, with nuisance parameters fitted to the data and refitted in each replicate; their calibration depends on that model and on the bootstrap approximation. Calibrated screening tests use the upper-tail probability $p = (1 + \#\{S_i \geq S_*\})/(n_{\text{cal}} + 1)$ with ties counted conservatively, and dataset scans report Benjamini–Hochberg adjusted q -values [22]. When calibration and test cases are statistically interchangeable (exchangeable), this rank-based p -value is valid at any sample size. AUROC intervals resample law families or mechanism groups. In the synthetic calibrated-detection evaluation of the extrapolation screen, the primary reported quantity is the true-positive

rate at a threshold targeted to 5% false positives on independent calibration, with the 1% threshold secondary; screening comparisons report AUROC. The 0.5 threshold on the extrapolation-screen percentile (the calibration median; Fig. 2) selects cases for further testing.

The assignment rule (Fig. S1) sends a dataset to the extrapolation screen, whose classification is descriptive, when an admitted form fits a contiguous fitted regime, otherwise to the named-prediction test when a pre-horizon prediction and error model are supplied, and otherwise to abstention; when a fitted regime exists and a pre-horizon prediction and error model are also supplied, both tests are run and reported, as for Millikan’s data. Nearly constant responses go to the named-prediction test, which tests a constant against noise, and point clouds with at least 20 observations also receive a learned score.

Specification timing differed across analyses. The historical comparisons are retrospective; the large-encoder temporal analysis was specified before its temporal scores were known, and later scope decisions and added comparisons are identified where they are reported (Notes S16 and 17).

4.3 Extrapolation screen

The extrapolation screen draws on a fixed vocabulary of power, exponential, linear and constant forms. Given a point cloud, a fixed rule takes the candidate fitted regime to be the contiguous stretch in which one admitted form fits with high R^2 , and the extrapolation region to be the complementary region. The search runs from both ends and fit quality decides whether the low- or the high-value direction is retained. The screen reports two statistics. The split-conformal nonconformity [29] is the mean absolute residual in the extrapolation region divided by the 90th percentile of the absolute residuals in the fitted regime, and the extrapolation-residual ratio is the median absolute log-residual in the extrapolation region divided by its value in the fitted regime. Each form is fitted from several initializations, and each denominator carries a floor, one percent of the median response magnitude in the fitted regime for the nonconformity and 0.01 for the median log-residual, so that a near-perfect fit cannot produce an unbounded ratio. Relations with fewer than 20 points are calibrated against size-matched simulations. The classical calibration curves are sampled from a larger point-cloud pool built from the same formula graph as the compact encoder’s corpus (below); the pool has no date metadata and contains at least 3.3% post-1900 records (Note S8).

4.4 Representation screen

The representation screen uses a neural encoder trained only on point clouds from pre-horizon relations, with no example of a failure and without any anomaly supervision; its learned score is the distance of a new cloud’s embedding from those of the training relations. At query time the encoder receives an unordered, normalized (x, y) cloud and nothing else: no equation, units, date, law name or post-cutoff label.

The large encoder, a permutation-invariant set transformer [30] aligned during training to a network that reads the formulas’ expression trees, is trained on point clouds derived from 3,441 dimensionless reductions of pre-1900 laws. Its formulas, like those of the compact encoder and the same-generator set, come from a machine-assembled formula graph, part of which was generated with language models or by combinatorial construction; pre-1900 status is set by the registry of dated sources, and the same-generator set (1,501 formulas, 96,064 point clouds) is produced from the same graph by the same pipeline, separated in time but without a complete dated registry. Held-out calibration and pre-1900 test formulas are separated by law family (Fig. S3), and grouping the test formulas by mechanism gives the nine pre-1900 control groups. The weights used for scoring are those selected by a rule fixed before any later-physics evaluation. Its learned score averages four ranks, each transformed by its own pre-1900 calibration empirical cumulative distribution function: cosine distances to the training formulas’ expression-tree embeddings, to their means over formulas treated as the same law and to mean point-cloud embeddings for each formula, and a shrinkage-Mahalanobis distance [31, 32] to the distribution of those mean point-cloud embeddings. Scores are summarized by the median over clouds within a formula and then within a law family or control group, and family- and group-level AUROCs use 2,000 grouped-bootstrap

replicates; unless stated otherwise, bracketed intervals in this paper are 95% bootstrap intervals resampled at that level. The untrained network is scored by the identical procedure. Architecture, training schedule and corpus sizes are given in Note S16.

To build the regenerated controls we regenerated each of the nine pre-1900 control groups from one representative formula reduction, using the 55-family benchmark’s point count, x -jitter, repeat count, noise range and heteroscedastic profile, and left the trained weights, calibration, score and aggregation unchanged. The same-generator set and the 1950-horizon extension are described in Note S16.

The compact encoder is trained on 205 pre-1900 point clouds from 40 families, without unit information. The formula expressions behind these point clouds were drawn from a machine-assembled formula graph, part of which was generated with language models or by combinatorial construction. A record was admitted only by exact name match to, or derivation ancestry from, a law family in a registry of dated pre-1900 sources, which supplies the family’s date and citation; the expressions themselves were not individually checked against the original source texts (Note S8). No synthetic departure example enters its training. Leave-family-out calibration holds out each of the 40 training families in full, and the threshold, the 95th percentile of the 40 held-out family scores, is fixed before evaluation. Scores are summarized by the median over eight resampled copies of each dataset and then over the datasets within a family. The generic-function control encoder shares the compact encoder’s architecture and training and is trained on 205 point clouds from smooth generic expressions in 40 pseudo-families matched to the historical set in dataset count, family-size profile and the elementary function types present. Architectures, training schedules and the retraining seeds are given in Note S8.

4.5 Post-1900 formula benchmark

The 55-family benchmark contains 55 law families first valid between 1901 and 1996, each carrying a primary historical citation; thermionic emission [33] is one example. Every law is represented by a one-dimensional form $y = f(x)$ in dimensionless variables over a fixed support, sampled eight times at 120 points with varying nuisance parameters, sampling and 0.5–2.5% noise, giving 440 datasets across 55 families. Special reductions, such as the Bardeen–Cooper–Schrieffer interpolation [34], the two-flavor neutrino survival curve [35] and the ideal quantum-Hall staircase [36], are listed in Note S15. No benchmark family overlaps a family of the pre-1900 corpus.

The compact encoder’s learned score is summarized by the median over resampled copies, datasets and families and compared with its fixed threshold. The extrapolation score reruns the full form selection and extrapolation procedure against the $n = 120$ thresholds of the size-matched classical calibration (785 simulations per size; below). Eleven cited pre-1900 families excluded from encoder training are the controls, AUROC intervals resample families, and twelve of the post-1900 laws deliberately overlap the affine, power or exponential forms. The eleven additional generated cases are described in Note S17.

4.6 Historical measurements and controls

We analyze the published COBE-FIRAS monopole spectrum with its pointwise errors, five Bertozzi values digitized from the 1964 report, NIST specific-heat reference polynomials for six materials, Kamerlingh Onnes resistance values including upper limits, Millikan sodium stopping potentials and Solar-System orbital constants. For each case we record the model set, fitted parameters, observation model, fitted regime, adaptive choices, decision rule and output. We evaluate null distributions under the pre-1900 and successor predictions on the same x -grid; explicit error floors stand in for unavailable covariance and for digitization uncertainty, and we vary them in sensitivity analysis.

To test whether the reduction in discrepancy is specific to the successor prediction, five wrong-shape forms, one per case (affine for FIRAS, quadratic through the origin for Bertozzi, a global power law for copper, a smooth quadratic for Kamerlingh Onnes and inverse frequency for Millikan), and five dimensionally coherent wrong-mechanism forms are refitted on the same case grids and observation models with 1,000 parametric-bootstrap replicates. The

381 wrong-mechanism forms substitute fermionic occupation for photon statistics, a proton for the
382 electron rest-energy scale, an Einstein oscillator [37] for Debye acoustic modes [21], Arrhenius
383 transport for the superconducting transition, and Compton recoil for the photoelectric slope.

384 Sixteen measured series were taken unmodified from the NIST Dataplot and Statistical
385 Reference Dataset collections; their selection, the specification of each series before scoring,
386 and the scoring and grouping procedure are described in Notes S14 and 18.

387 4.7 Synthetic suites, baselines and localization

388 The baseline comparison uses 36 synthetic datasets, on which the extrapolation screen is
389 compared with its two statistics. Cumulative-sum (CUSUM) change-point detection [3]
390 and Ramsey’s regression specification error test (RESET) [38] are compared with the
391 extrapolation-residual ratio on a separate suite of 16 phenomena (Note S3; Fig. S2d–f). The
392 mechanism-level benchmark contains 12 classical and 12 synthetic departure families with
393 eight datasets each; its 5% and 1% thresholds come from 785 valid size-matched calibration
394 simulations, and uncertainty is resampled by departure or classical family. The compact
395 encoder is evaluated on 192 datasets over the same 24 families. Instrument tests apply
396 calibration drift, sensor saturation, censoring, x -axis error, outliers, correlated noise and
397 heteroscedasticity to classical synthetic relations while leaving the underlying law intact, and
398 boundary intervals are obtained by split-conformal calibration of the normalized localization
399 error separately at $n = 40, 80$ and 160 .

400 5 Data availability

401 COBE-FIRAS data are from NASA’s Legacy Archive for Microwave Background Data Analy-
402 sis; Bertozzi’s, Millikan’s and Kamerlingh Onnes’s values are digitized from Refs. [13, 17, 15];
403 specific-heat curves are from the NIST Cryogenic Material Properties database; the 16
404 measured series are from the NIST Dataplot and Statistical Reference Datasets collections;
405 Solar-System and Galilean-moon constants and Boyle’s table are from standard references
406 (Note S10). The data package contains source URLs and hashes; the 55-family benchmark and
407 its 440 generated point clouds; the 11 pre-1900 control families and their 88 generated datasets;
408 all 128 individual scores of the measured series; the pre-1900 training, calibration and evalua-
409 tion manifests; and every numerical result used in the figures. The project repository is avail-
410 able at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>,
411 and the release used here will be archived on Zenodo on acceptance under the concept DOI
412 <https://doi.org/10.5281/zenodo.22138620>, which resolves to the most recent version.

413 6 Code availability

414 Code is available at [https://github.com/ChenxiHeCam/](https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons)
415 [Epistemic-probe-of-Classical-Horizons](https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons). The repository contains corpus checks,
416 point-cloud generators, evaluators, statistical calibration and robustness experiments, model
417 definitions, trained encoder weights, training configurations, figure scripts and the validation
418 scripts. The release tagged `epoch-manuscript-2026-10-04` contains the code and result
419 files used for the numbers reported here.

420 7 Use of AI tools

421 Part of the formula graph from which both encoders’ training formulas and the same-
422 generator set were drawn was generated with language models (Claude Haiku, Anthropic;
423 DeepSeek) (Methods). Claude (Anthropic) and Codex (OpenAI) were used as assistants for
424 code development, figure scripting and language editing. The authors designed the study,
425 checked every analysis, figure and sentence, and take full responsibility for the content.

8 Acknowledgments 426

We thank NASA’s Legacy Archive for Microwave Background Data Analysis (LAMBDA), 427
the NIST Cryogenic Material Properties database and the digital archive of the Royal 428
Netherlands Academy of Arts and Sciences (KNAW) for public data access. This research 429
received no specific grant from any funding agency in the public, commercial, or not-for-profit 430
sectors. 431

9 Author contributions 432

C.H. conceived the framework, designed and built EPOCH, ran the experiments, and wrote 433
the manuscript. J.C. contributed to the benchmark construction, case-study analysis, and 434
manuscript revision. Both authors reviewed and approved the final manuscript. 435

10 Competing interests 436

The authors declare no competing interests. 437

Correspondence and requests for materials should be addressed to C.H. 438
(ch2067@cam.ac.uk). 439

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